

**ACML101: Introduction to
Chemistry
Inorganic Chemistry: Tutorial 2
Coordination Chemistry**

(1) Jahn-Teller effect is not observed in high spin octahedral complexes of

- (A) d^7 (B) d^8 (C) d^4 (D) d^9

(2) (a) The value of Δ_o in $[\text{Mn}(\text{OH}_2)_6]^{3+}$ is $15,800 \text{ cm}^{-1}$ while the mean pairing energy (P) in this complex is $28,000 \text{ cm}^{-1}$. Do you expect this ion to be high or low spin? (b) Determine whether either spin state for this complex is likely to be subject to a Jahn-Teller distortion. If it can occur, which spin state would display the greater effect?

Q3. An aqueous solution of $[\text{TiCl}_6]^{3-}$ showed an absorbance maxima at 13000 cm^{-1} with a broad hump at 12500 cm^{-1} in the UV-vis spectrum. Explain the origin of these transitions.

(4) The value of ϵ_{max} in the electronic spectra of $[\text{CuCl}_4]^{2-}$ and $[\text{Cu}(\text{OH}_2)_6]^{2+}$ differ by a factor of about 100. Comment on this observation and state which complex you expect to exhibit the larger value of ϵ_{max} . (previous year question)

(5) Arrange the following species in the order of increasing Jahn Teller distortion in an octahedral environment high-spin cobalt (III) and low-spin Co(II)

(6) Which complex is more stable $\text{cis}[\text{Cu}(\text{en})_2 2\text{H}_2\text{O}]$ or $\text{trans}[\text{Cu}(\text{en})_2 2\text{H}_2\text{O}]$

7. Find out the correct spinel structure for FeCr_2O_4 and NiAl_2O_4

8. Why MgAl_2O_4 form normal spinel ?

9. Solutions of $[\text{Cr}(\text{OH}_2)_6]^{3+}$ ions are pale blue green but the chromate ion, CrO_4^{2-} , is an intense yellow. Characterize the origins of the transitions and explain the relative intensities.

10. Following is the electronic absorption spectra of an aqueous solution of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ (solid line). It has been observed that the spectrum is blue-shifted (towards higher energy) over time upon keeping the solution for hours, as indicated by short dashes and dot-dashes. Explain the observation based on your knowledge about crystal field theory.

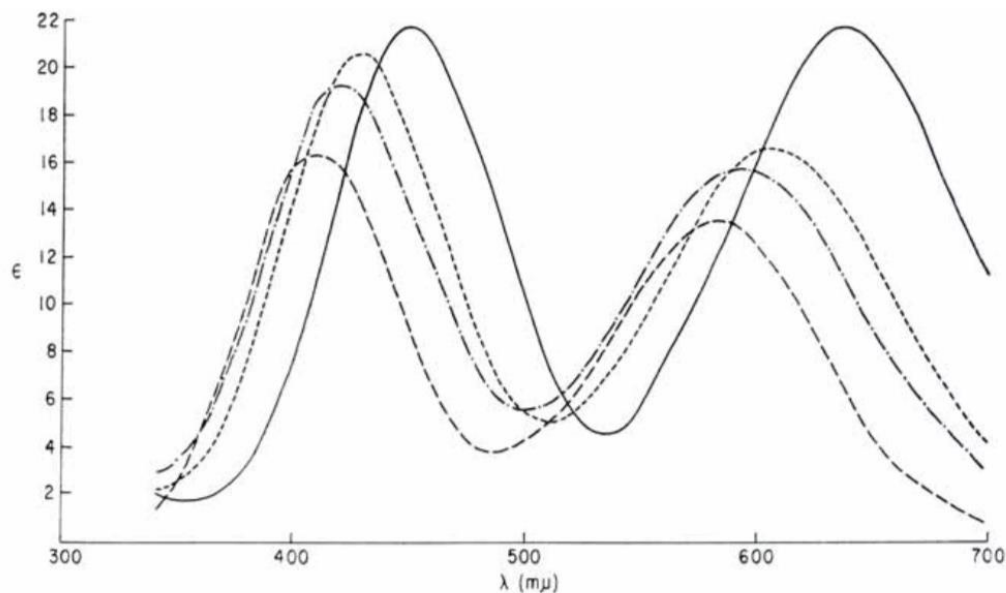


Figure 4. The solid line represents absorption spectrum of an aqueous solution of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ immediately after preparation; the short dashes represent the same solution after two hours; the dot-dash line represents the solution after 6.5 hours; and the long dashes represent the spectrum of $\text{Cr}(\text{H}_2\text{O})_6(\text{NO}_3)_3 \cdot 3\text{H}_2\text{O}$.

Q11. An aqueous solution of a 0.5 M solution of manganese chloride is very weakly colored, although an aqueous solution of potassium permanganate of same concentration exhibits very intense color. Explain.

Q12. No *d* electrons are present in Cr in $\text{K}_2\text{Cr}_2\text{O}_7$ and K_2CrO_4 . However, these two compounds are intensely colored when dissolved in water. Explain.

Additional questions which are out of syllabus

(1) Determine how the p orbitals would split in an (a) octahedral and (b) trigonal bipyramidal crystal field.

(2) Determine the d orbital splitting to be expected (qualitatively) in a T-shaped crystal field (i.e. for a square planar arrangement missing one ligand). Take the complex plane to be the xy plane and make sure you clearly distinguish between the x and y directions. *NOTE: represent your answer as a qualitative energy level diagram.*